

Bank Interest Rate System

Title: Implementation of method overriding using Bank Interest program

Aim : To develop a Java program to calculate and display the rate of interest offered by different banks using inheritance, method overriding, and runtime polymorphism.

Software requirements:

Software : Java JDK , VS Code / Eclipse IDE, Windows Operating System

Hardware : Computer/Laptop, Minimum 4 GB RAM, Keyboard , Monitor

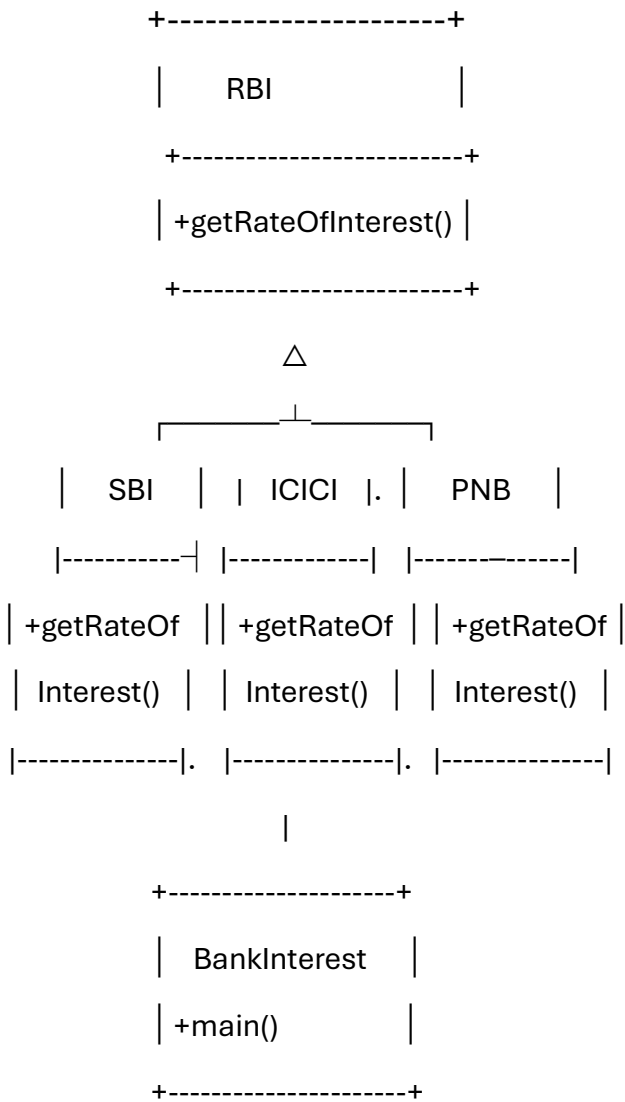
Algorithm:

1. Start the program.
2. Import the Scanner class.
3. Create a base class named RBI.
4. Define the getRateOfInterest() method in the RBI class and return 4.0%.
5. Create the SBI class by extending the RBI class.
6. Override the getRateOfInterest() method in the SBI class and return 7.0%.
7. Create the ICICI class by extending the RBI class.
8. Override the getRateOfInterest() method in the ICICI class and return 6.5%.
9. Create the PNB class by extending the RBI class.
10. Override the getRateOfInterest() method in the PNB class and return 6.0%.
11. Read the bank name from the user.
12. Create an RBI reference.
13. Based on the entered bank name, create the corresponding bank object.
14. Call the getRateOfInterest() method using the RBI reference.
15. Display the rate of interest.

16.Display an error message if an invalid bank name is entered.

17 . Stop the program.

UML Diagram :



Source Code :

```
import java.util.Scanner;

class RBI {

    // Returns the base RBI rate of interest

    double getRateOfInterest() {

        return 4.0;

    }

}

class SBI extends RBI {

    // Overrides the RBI interest rate

    @Override

    double getRateOfInterest() {

        return 7.0;

    }

}

class ICICI extends RBI {

    // Overrides the RBI interest rate

    @Override

    double getRateOfInterest() {

        return 6.5;

    }

}

class PNB extends RBI {

    // Overrides the RBI interest rate

    @Override

    double getRateOfInterest() {
```

```

        return 6.0;
    }
}

public class BankInterest {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the Bank name to find the rate of Interest:");

        String bank = sc.nextLine();

        RBI rbi;

        // Select the bank object based on user input

        if (bank.equalsIgnoreCase("RBI")) {

            rbi = new RBI();

        } else if (bank.equalsIgnoreCase("SBI")) {

            rbi = new SBI();

        } else if (bank.equalsIgnoreCase("ICICI")) {

            rbi = new ICICI();

        } else if (bank.equalsIgnoreCase("PNB")) {

            rbi = new PNB();

        } else {

            System.out.println("Invalid Bank Name");

            sc.close();

            return;

        }

        System.out.println("RBI rate of interest is : " + rbi.getRateOfInterest() + "%");

        sc.close();

    }
}

```

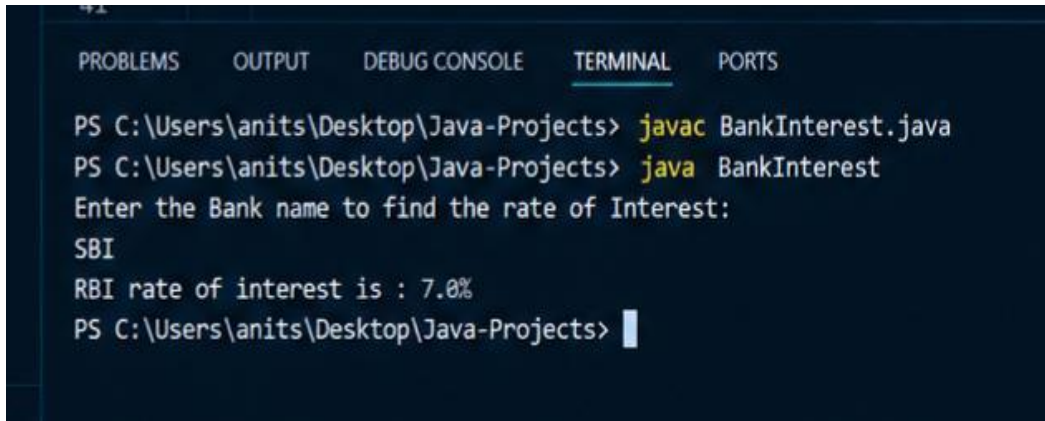
```
}
```

Compilation & Execution Commands :

Compilation: `javac BankInterest.java`

Execution: `java BankInterest`

Output screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\anits\Desktop\Java-Projects> javac BankInterest.java
PS C:\Users\anits\Desktop\Java-Projects> java BankInterest
Enter the Bank name to find the rate of Interest:
SBI
RBI rate of interest is : 7.0%
PS C:\Users\anits\Desktop\Java-Projects> 
```

Result : The Bank Interest Rate System was successfully implemented in Java using inheritance, method overriding, and runtime polymorphism. The program successfully displays the rate of interest for RBI, SBI, ICICI, and PNB based on the bank name entered by the user.